SWARMRIDE BHARAT · SELF-AWARE TRAINS FOR INDIAN RAILWAYS

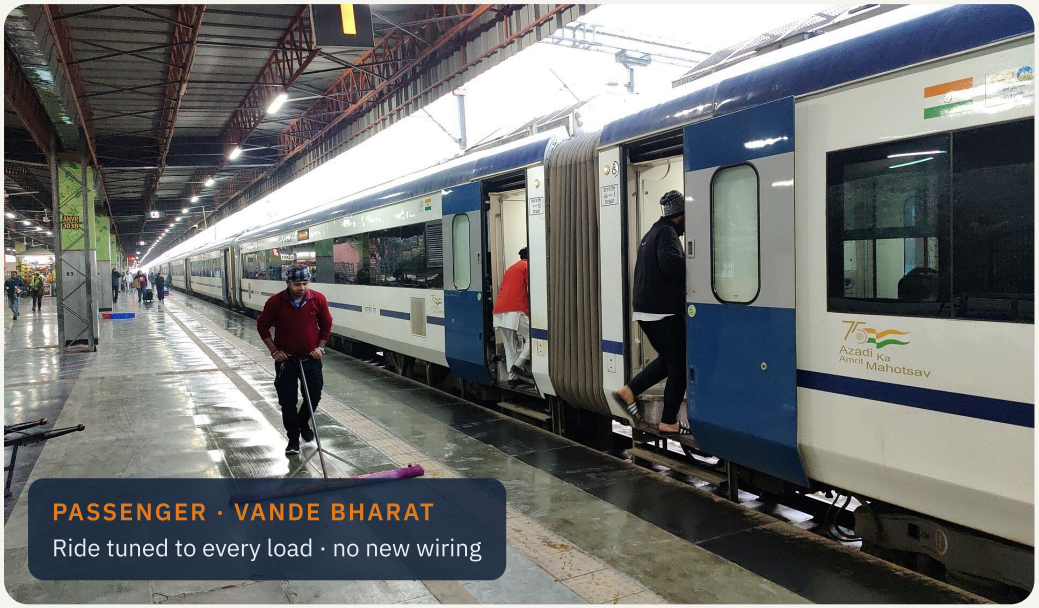
# Every bogie powers itself, rides smarter and checks itself.

## WHAT WE DO

We replace the oil damper on each bogie with a **smart magnet-and-coil damper**. The bump energy that becomes heat today now **powers safety sensors** and **tunes the ride**.

## APPROACH

1. **Harvest** — Smart damper makes power
2. **Store** — Power bank on every bogie
3. **Sense + tune** — Axles, track, load, ride
4. **Sync** — Loco warns every coach




PASSENGER · VANDE BHARAT
Ride tuned to every load · no new wiring




GOODS · BOXN WAGONS
Every axle watched · wagons power themselves

**~3 / day**
hot-axle cases on Indian Railways
Railway Board, 2025

**327**
derailments from track faults, 2017–21
CAG audit

**30–100%**
shortfall in Track Recording Car checks
CAG audit

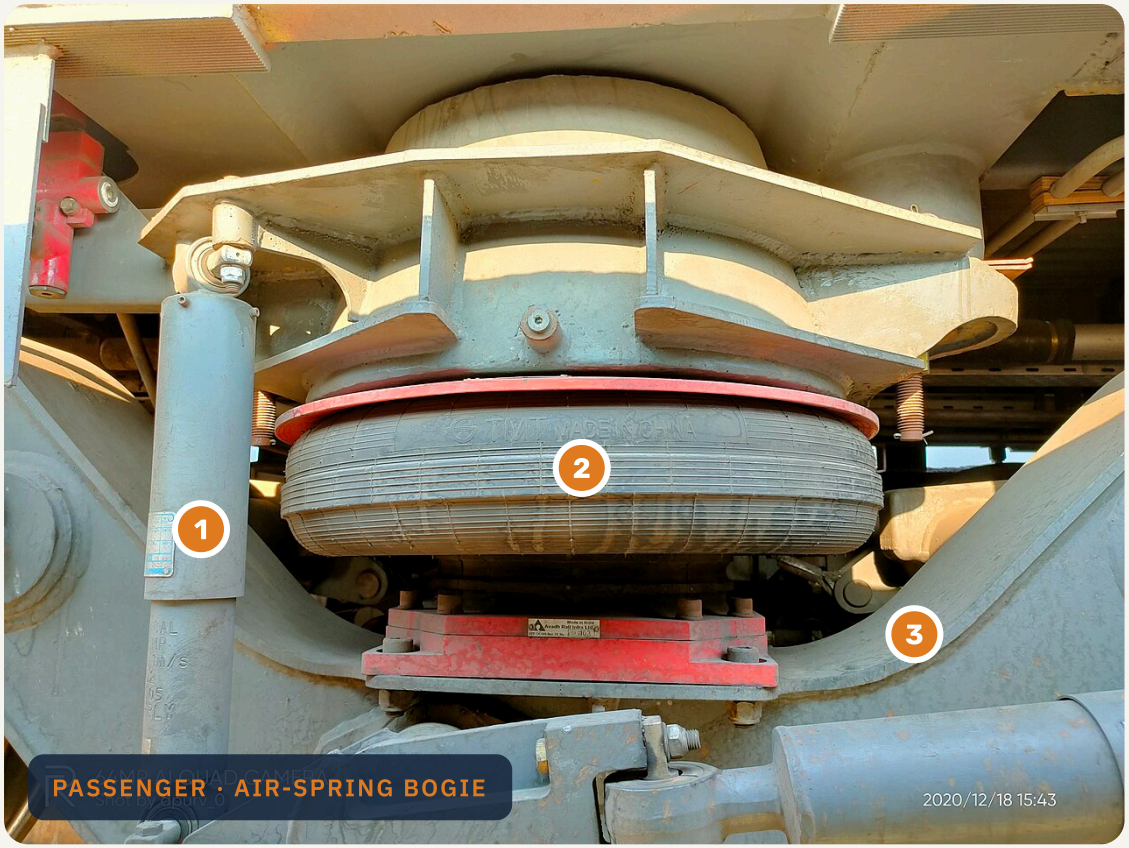
**0 W**
electrical power on a goods wagon today
no sensors possible

**The energy is already in every bump. Today it becomes heat. We turn it into safety and comfort.**

Photos (Wikimedia Commons): Vande Bharat Express — Ravi Dwivedi, CC BY-SA 4.0 · BOXN freight train at Rajahmundry — Adityamadhav83, CC BY-SA 3.0

WHERE IT FITS · REAL BOGIES, REAL PARTS

# Same bogie. We swap one part and add a small power bank.




1 **Oil damper today → SwarmRide smart damper.** Same mounts. Moves with the suspension and makes power.

2 **Air spring stays.** Its pressure gives the live passenger load.

3 **Bogie frame → sealed power bank + controller.**




4 **Secondary springs + oil damper** → smart damper fits here. 5 **Axle box** → bearing sensor.

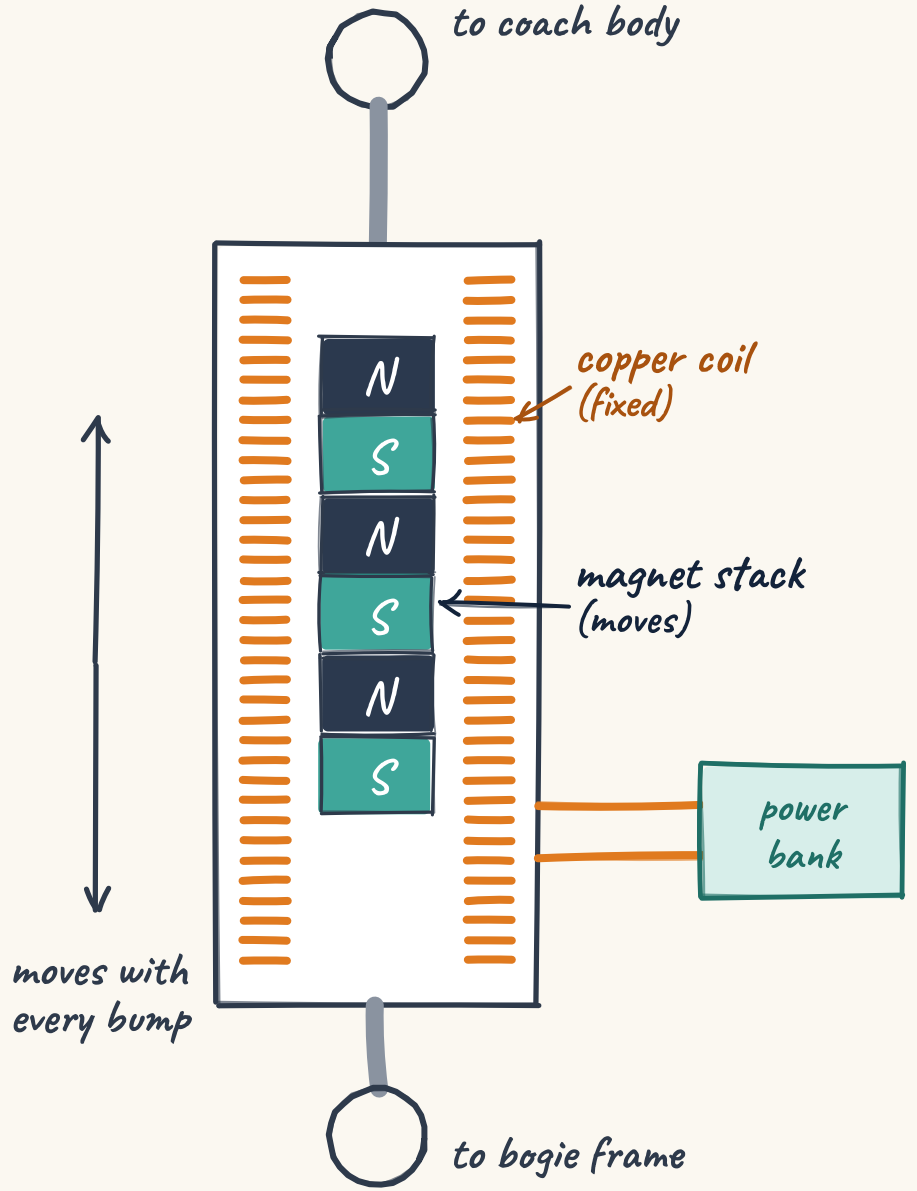



6 **Spring nest** → compact smart damper beside it. 7 **Axle box** → bearing temperature + vibration sensor.

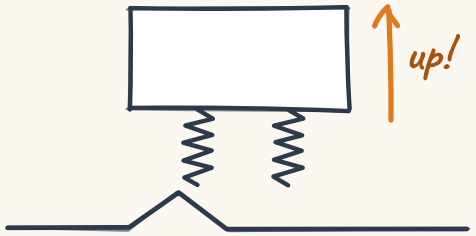
**No new bogie • no new cables • fits today's mounts**

Photos: air spring on LHB coach — Apurv013, CC0 (Wikimedia Commons) · ICF coach bogie and Casnub wagon bogie — supplied by the author

# A magnet moving through a coil: power and ride control in one part.

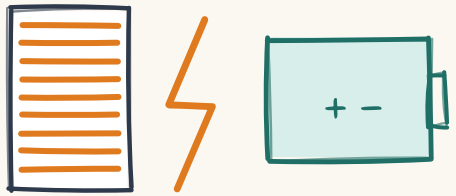



## 1 · A bump moves it



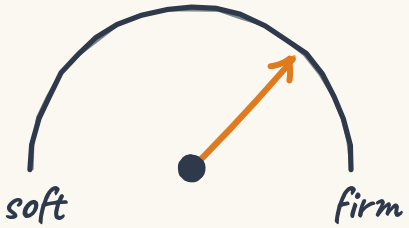

Track bump → coach moves → magnet slides through the coil

## 2 · It makes power



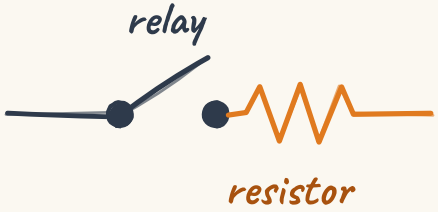

**e = Ke · v** (Faraday's law) → charges the bogie power bank

## 3 · It sets the ride




**c = Ke² ÷ (R coil + R load)** → firmness set in milliseconds

## 4 · It fails safe

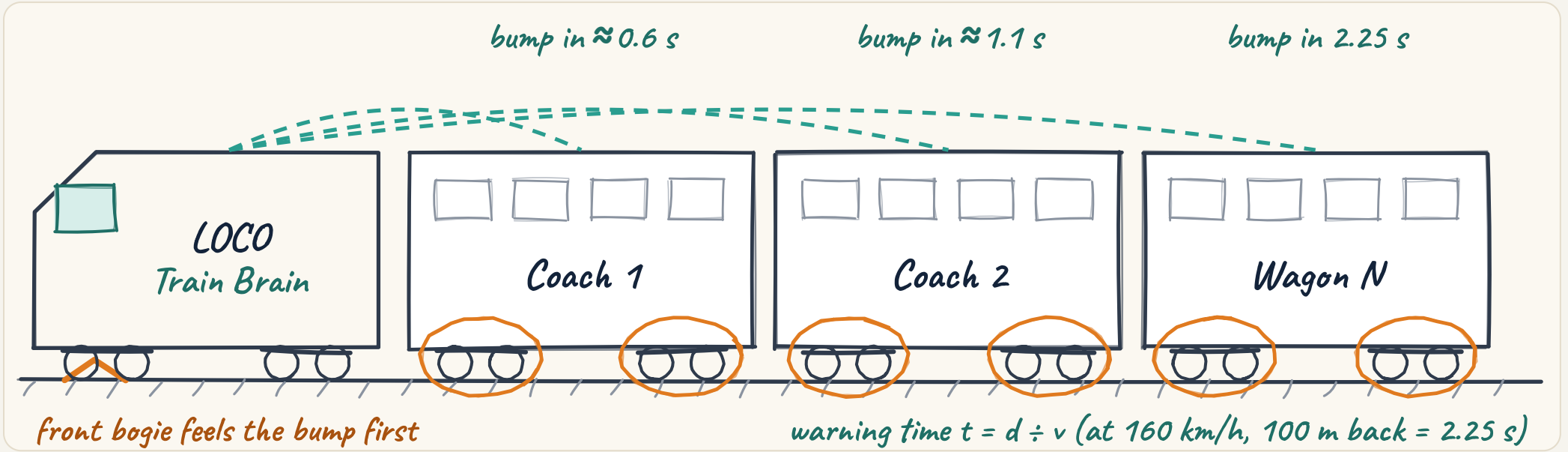



Power lost → relay → resistor → works like a normal damper

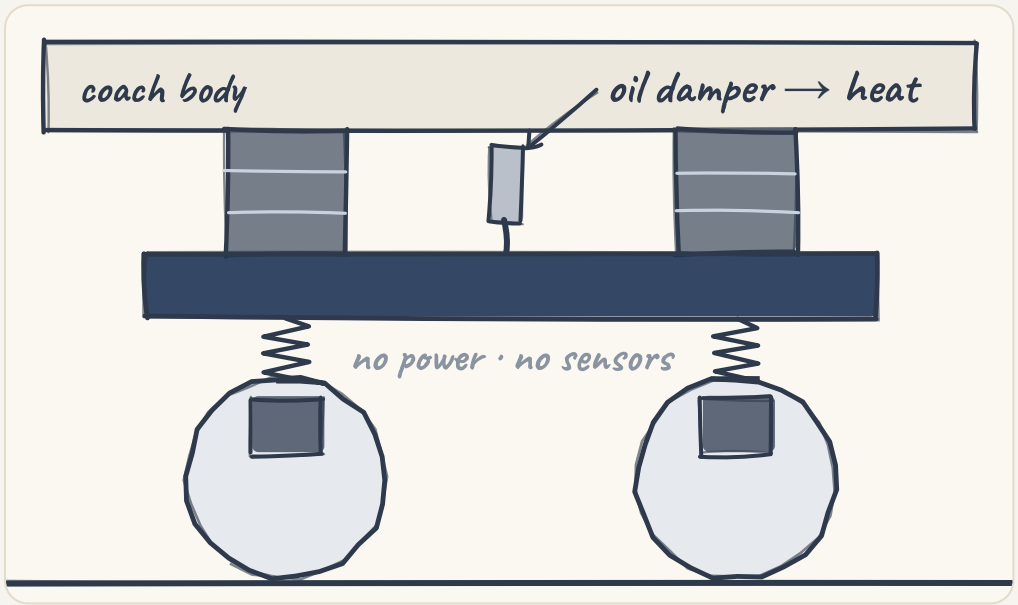
**Fits existing damper mounts • no oil • no gears**

Concept sketches, not to scale

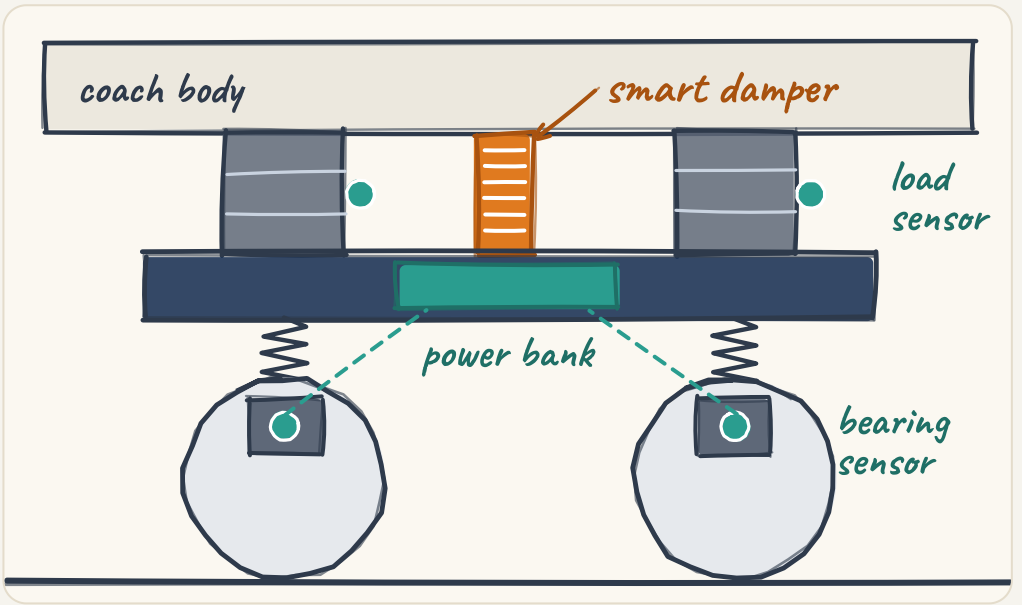
# The loco warns every coach before the bump arrives.




## Bogie today




## SwarmRide bogie




Orange rings: SwarmRide on every coach and wagon bogie · concept sketches, not to scale

TWO PRODUCTS · ONE TECHNOLOGY

# Comfort for passenger trains. Safety for goods trains.

## SwarmRide Passenger

Vande Bharat · LHB · ICF




- Ride tuned to live load from air-spring pressure
- Ready before each bump: warning from the loco
- Track recorded on every trip
- No new wiring: each bogie powers itself

## SwarmRide Freight

BOXN · DFC rakes




- Every axle: bearing temperature + vibration
- Train parting found at the exact wagon
- Track recorded on every goods run
- Wagons have no power; now they make their own

| RIDE | HOT AXLES | PARTING | TRACK CHECKS | BOGIE POWER |
|---|---|---|---|---|
| one load | ~25 checkpoints | last wagon | periodic | none |
| ↓ | ↓ | ↓ | ↓ | ↓ |
| **every load** | **every axle, always** | **exact wagon** | **every train, daily** | **self-powered** |

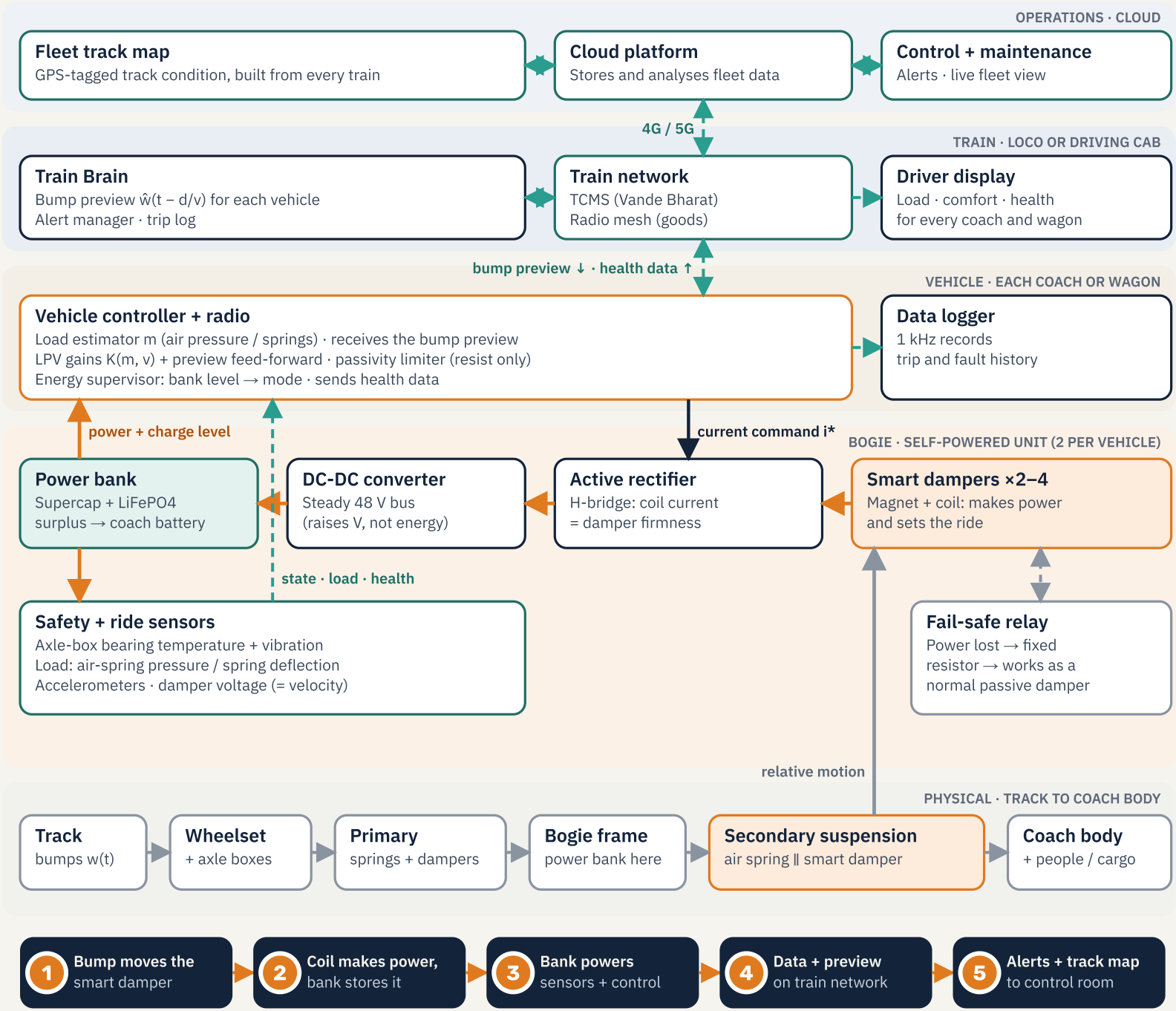
**Prove it · simulation + bench** → **RDSO approval · trials** → **Pilots · DFC rake + Vande Bharat** → **Fleet rollout · factories + data**

## One damper. Two markets. Every bogie powers itself.

Photos: ICF coaches — supplied by the author · BOXN freight train — Adityamadhav83, CC BY-SA 3.0 (Wikimedia Commons) · expected results, to be proven

END-TO-END ARCHITECTURE

# From a bump on the track to the control room.




## LEGEND

- Harvested power
- Data link (arrows show direction)
- Control command
- Mechanical motion

## ENERGY ORDER ON EVERY BOGIE

1. Safety sensors
2. Radio link
3. Ride control
4. Reserve for stops
5. Surplus to coach battery

## CONTROL LAW

$F^* = -K(m,v)\cdot\hat{x} + Kp\cdot\hat{w}$

$c = \text{clip}(F^* \div v_{rel}, 0, c_{max})$

$i^* = c\cdot v_{rel} \div Kf$

Resists only: every bit of damping is harvested energy.

## WORKS ALONE

If the train network drops, each vehicle keeps load-adaptive control without the preview.

## SAFETY

Fail-safe relay makes it a normal damper. Safety never depends on software. To be tested to EN 50155 · EN 61373 · EN 50121.